

TWO Disciplinary power

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Michel Foucault's *Discipline and Punish: The Birth of the Prison*, published in 1975, contains his most famous and elaborate exposition of disciplinary power. A bird's-eye view of his preceding and succeeding analyses reveals, however, that this concept arose in overlapping stages and served a variety of purposes. From roughly 1973 to 1976, in analyses of punishment, proto-psychiatry, criminology and race war, Foucault attempted to articulate disciplinary power in contradistinction to sovereign power. From about 1976 to 1979, he used disciplinary power as a springboard for delineating modalities of power concerned with population, namely, biopolitics, security and governmentality. Finally, in the early 1980s disciplinary power figured more as an implicit background to his analyses of subjectivity in Greco-Roman antiquity and early Christianity. The long shadow cast by this concept renders it absolutely crucial to understanding the trajectory of Foucault's thought.

Using a composite account of disciplinary power drawn from Foucault's seminal presentation in *Discipline and Punish* as well as his Collège de France course for the academic year 1973–74, *Psychiatric Power*, I will provide an overview of disciplinary power and then exemplify the exercise of this power through Frederick Winslow Taylor's *The Principles of Scientific Management*, published in 1911. Taylor's *Principles*, which influenced American and European industrialists as well as Lenin and Antonio Gramsci, enriches our understanding of disciplinary power in two ways. First, the presentation of scientific management at the core of *Principles* reflects nothing short of a full-fledged disciplinary programme. Indeed, it is hard to read a page of *Principles* without noticing that Taylor suffuses his presentation with a thoroughly disciplinary aspiration. Second, Taylor's *Principles* highlights the limitations to the exercise of disciplinary power by attesting that disciplinary practices bound up with the application of scientific management are deeply contested. Taylor thus de-naturalizes this form of power even as he seeks to extend its reach not only within factories, but also within "all social activities", including the management of homes, farms, businesses, churches, charities, universities and governmental agencies (F. W. [Taylor 1967](#): 8).

Subjected individuals

The concept of disciplinary power concerns individuals. As Foucault notes with reference to what he takes to be the ideal exercise of this power, "We are never dealing with a mass, with a group, or even, to tell the truth, with a multiplicity: *we are only ever dealing with individuals*" (2006a: 75, emphasis added). However, in opposition to political theories which take the individual as a given for the purpose of constructing sovereignty, as in the notable case of Thomas Hobbes's version of the social contract, Foucault sets about showing that the individual first and foremost amounts to a construction of disciplinary power. The individual is an effect of this form of power rather than the raw material upon

which it impinges. Foucault writes, “Discipline ‘makes’ individuals; it is the specific techniques of a power that regards individuals as objects and as instruments of its exercise” (1979: 170). As a first approximation, we can therefore say that disciplinary power produces individuals as its objects, objectives and instruments.

Disciplinary power yields such effects by targeting bodies. The targeting of bodies may not seem terribly unique, especially in light of Foucault’s sweeping assertion that “what is *essential* in *all* power is that *ultimately* its point of application is *always* the body” (2006a: 14, emphasis added). It seems even less singular in light of his suggestion that pastoral power treats the body as an object of care (2007: 126–8) and that even sovereign power sets its sights on the body as an object of violence or honour (2006a: 44–5). However, what distinguishes disciplinary power from these other modalities of power is its endeavour to meticulously, exhaustively and continuously control the activities of bodies so as to constitute them as bearers of a highly particular relationship between utility and docility, whereby increases in utility correspond to increases in docility and vice versa. In Foucault’s words, disciplinary power strives to make the body “more obedient as it becomes more useful, and conversely” (1979: 138). This increase entails the augmentation of the skills and aptitudes of bodies without at the same time allowing these skills and aptitudes to serve as a source of resistance to disciplinary power. This form of power thereby attempts to resolve the problem of the resistances aroused from its own incessant investments in the body. Disciplinary power controls the body to effectuate this result through the production not only of an individual but also of individuality, the amalgam of qualities that render an individual distinct from others (Arendt 1985: 454). This individuality consists of cellular, organic, genetic and combinatory traits. Let us now outline the production of these traits.

Foucault insists that disciplinary power creates a cellular form of individuality by ordering individuals in space. He calls this ordering “the art of distributions”. Cellular individuality rests on the division of individuals from others. The art of distributions produces this individuality by first of all enclosing a space different from all others through the use of walls or gates, as in the case of barracks and factories (Foucault 1979: 141–3). It partitions this space into individual cells in order to break up collective activities that deter from the goal of utility, such as desertion or vagabondage. The art of distributions also codes a space with specific functions to make it as useful as possible (*ibid.*: 143–5). As an example of this coding, Foucault refers to the production of printed fabrics at the Oberkampf manufactory at Jouy. The workshops at the manufactory were divided into operations “for the printers, the handlers, the colourists, the women who touched up the design, the engravers, the dyers” (*ibid.*: 145). Each worker occupied a space defined by his or her specific function within the overall production process. Lastly, the art of distributions creates a cellular individuality by ascribing the unit of rank to individuals. As an example of rank, Foucault discusses the seating of pupils in a classroom according to their age, grade and behaviour (*ibid.*: 146–7).

Within this enclosed space, disciplinary power produces an organic individuality by exerting a control over bodily activities. This individuality is “organic” in so far as it lends itself to disciplinary practices all on its own, as if spontaneously and naturally (*ibid.*: 155–6). The control of bodily activities realizes this organic individuality first of all through a temporal enclosure afforded by the use of timetables, which prevent idleness by partitioning activities into minutes and seconds (*ibid.*: 150–51). The control of

activities also breaks down movements of the body into an ever-greater number of acts and indexes these acts to temporal imperatives. Foucault identifies the prescription of the duration and length of the steps of marching soldiers as an example of this temporal elaboration of the act (*ibid.*: 151–2). The control of activities further implies a relationship between the general position of the body and its gestures. In this regard, Foucault mentions the example of the upright posture of pupils and the correct positioning of their elbows, chins, hands, legs, fingers and stomachs as the conditions for good handwriting (*ibid.*). The control of activities goes even further, correlating the gestures of the body to the parts of the object used by it, as in the case of manifold gestures employed by a soldier to manipulate the barrel, butt, trigger-guard, notch, moulding, lock, screw and hammer of a rifle (*ibid.*: 153). Finally, rather than merely preventing idleness, the control of activities forges an organic individuality by exhaustively using time.

With the activities of the body controlled, disciplinary power proceeds to constitute a genetic form of individuality by subjecting the body to the demand for a perpetual progress towards an optimal end. Foucault dubs this demand the “organization of geneses”. Drawing from the example of the military, he submits that perpetual progress towards an end yields a genetic individuality in the following ways: first, through the division of time into distinct segments, such as periods of practice and training; second, through the organization of these segments into a plan proceeding from the simplest elements, such as the positioning of the fingers in military exercise; third, through the ascription of an end to these segments in the form of an exam; and, finally, through the production of a series that assigns exercises to each individual according to rank (*ibid.*: 157–9).

Finally, disciplinary power establishes a combinatory form of individuality characterized by articulations with other bodies to obtain a level of efficiency greater than that realized by the mere sum of the activities of these bodies (*ibid.*: 167). Foucault calls this process the composition of forces. This composition gives rise to a combinatory individuality by first treating individual bodies as mobile elements to be connected to other individual bodies as well as the totality of bodies; second, by coordinating the time of each of these bodies to maximize the extraction of their forces and to combine them with others for the optimal results; and, lastly, by commands that may be transmitted through signs and that therefore need not be verbalized, much less explained (*ibid.*: 164–7).

We now know *how* disciplinary power works and *what* it produces. It works by distributing individuals, controlling activities, organizing geneses and composing forces, and these functions correspond to the production of cellular, organic, genetic and combinatory individualities, respectively. Yet, at the risk of drawing too fine a distinction, Foucault goes further in his analysis to impart a sharp sense of how disciplinary power *gets going* and *keeps going*. He attributes the success of this power to several basic techniques: hierarchical observation, normalizing judgement and the examination.

If architecture figures within the art of distributions as a means of ordering multiplicities into cellular individuals, it plays the role within hierarchical observation of rendering individuals visible with the overall effect of structuring their behaviour. In making individuals seeable, architecture serves, as Foucault writes, “to act on those it shelters, to provide a hold on their conduct, to carry the effects of power right to them, to make it possible to know them, to alter them” (*ibid.*: 172). Still, he suggests that outside of any ideal schema, architecture alone falls short of making visibility constant. What makes this visibility perpetual is the implementation of a hierarchical network within the group of individuals who

occupy a particular architectural space. Foucault offers many examples of these networks. While we will not dwell on them here (we have an opportunity to gauge their presence in Taylor's *Principles*), it is instructive to mention briefly one particularly rich example. Foucault suggests that surveillance operated in the asylum in the early nineteenth century not only through a doctor but also through supervisors who reported on patients, and servants who feigned servitude to patients while gathering and transmitting information about them to the doctor (2006a: 4–6). This example clearly demonstrates the communication of the gaze from the top to the bottom, its manifestly “hierarchical” character. Yet, Foucault is keen to remind us that the gaze may operate in a more multi-directional manner to the point of bearing on the supervisors themselves. Such a dense network of vigilant and multi-directional gazes no doubt causes disciplinary power to appear ubiquitous, but the sheer simplicity of its mechanism also makes it seem rather inconspicuous (*ibid.*: 177).

Although surveillance rests on individuals, its functioning is that of a network of relations from top to bottom, but also to a certain extent from bottom to top and laterally; this network “holds” the whole together and traverses it in its entirety with effects of power that derive from one another: supervisors, perpetually supervised.

(1979: 176–7)

In a disciplinary world, however, it is not enough to see bodies so as to yield from them specific effects. One must be able to *judge* them as well. This modality of power therefore depends on normalizing judgement for its continued exercise. Foucault indicates that this form of judgement consists of features that make it look quite different from judgement in, for example, criminal courts. These features are summed up in terms of the following forms of punishment: first, even minute departures from correct behaviour are punished; second, failure to adhere to rules established on the basis of regularities observed over time is punished; third, exercise is used specifically as a corrective punishment; fourth, gratification is used in addition to punishment for the purposes of establishing a hierarchy of good and bad subjects; and, finally, rank understood as the place occupied in this hierarchy is used as a form punishment or reward (*ibid.*: 177–83). What ultimately stands out here for Foucault is the concept of the norm. Disciplinary power judges according to the norm. By “norm”, however, it should be obvious that Foucault has in mind something other than a strictly legal concept. He depicts the norm as a standard of behaviour that allows for the measurement of forms of behaviour as “normal” or “abnormal”. In his words, “the norm introduces, as a useful imperative and as a result of measurement, all the shading of individual differences”. The norm thus establishes the figure of the “normal” as a “principle of coercion” for the figure of the “abnormal” (*ibid.*: 184).

The examination combines the techniques of hierarchical observation and normalizing judgement in “a normalizing gaze” to lend further sustenance to the exercise of disciplinary power (*ibid.*). This gaze, as Foucault points out in a splendidly economical formula, “manifests the subjection of those who are perceived as objects and the objectification of those who are subjected” (*ibid.*: 184–5). Put differently, the examination binds the exercise of disciplinary power to the formation of a disciplinary knowledge. It does so in several ways. First of all, the examination facilitates the exercise of disciplinary power by objectifying subjects through observation. As Foucault posits, “Disciplinary power manifests its

potency, essentially, by arranging objects. The examination is, as it were, the ceremony of this objectification” (*ibid.*: 187). In this regard, he mentions the first military review of Louis XIV as a form of examination yielding the objectification of subjects. This review subjected 18,000 soldiers to the gaze of a barely visible sovereign who commanded their exercises (*ibid.*: 188). Second, the examination constitutes individuality through an administrative form of writing that leaves behind a dense layer of documents, as in the examples of medical records and student records. This writing makes it possible to describe individuals as objects and track their development, or lack thereof, as well as to monitor through comparison phenomena within the larger aggregate of population (*ibid.*: 189–91). Finally, the accumulation of documents through the examination forges the individual as a case defined in terms of a status bound up with all of the “measurements”, “gaps” and “marks” characteristic of disciplinary power (*ibid.*: 192).

In historical terms, Foucault sketches the shift from a society (prior to the sixteenth century) in which disciplinary power played a marginal but critical and innovative role from within the confines of religious communities to a society (beginning in the eighteenth century) in which it played a preponderant role from a myriad of institutions. In this sketch, disciplinary power spread initially through several “points of support” (2006a: 66) with religious underpinnings, such as the education of youth inspired by the ascetic ideal embraced by the Brethren of the Common Life with its focus on progressive stages of education, rules of seclusion, submission to a guide and military organization; colonization as practised by the Jesuits in the Guarani republic of Paraguay with its emphasis on the full employment of time, permanent supervision and the cellular constitution of families; and, lastly, the confinement of marginal elements of the population under the management of religious orders. From these peripheral positions, disciplinary power began to cover more spheres of society without any religious backing, appearing in the army by the end of the seventeenth century and working class by the eighteenth century (*ibid.*: 66–71).

Foucault maintains that this formidable extension of disciplinary power across the surface of society reflected a deeper ensemble of transformations. First, disciplinary power began to function as a technique more for the constitution of useful individuals than for the prevention of desertion, idleness, theft and other problems. Second, disciplinary mechanisms began to extend beyond their institutional parameters to yield lateral effects. In this regard, Foucault mentions the quite fascinating example of schools using information gathered from students to monitor parental behaviour. Lastly, disciplinary power began to bear on society as a whole through the organization of a police apparatus concerned with intricacies of individual behaviour (1979: 210–16).

These transformations were bound up in their turn with broad historical processes in economic, juridical and scientific domains. The generalization of disciplinary power took place against the background of the eighteenth-century problem of indexing the rapid growth in population to the rapid growth in production apparatuses (*ibid.*: 218–20). It attempted to resolve this problem by offering a means of administering the growth in the number of human beings and making them useful. The generalization of disciplinary power also entailed consequences for the juridical system, introducing asymmetries that vitiated the egalitarian juridical framework forged in the eighteenth century. As Foucault explains, disciplinary power established relationships of constraint between individuals rather

than relationships of contractual obligation, and it defined individuals hierarchically rather than universally. The play of such asymmetries within the time and space proper to the exercise of disciplinary power effectively suspended the law (*ibid.*: 222–3). Lastly, the generalization of disciplinary power implied a tightening of relations between power and knowledge to the point of their mutual constitution by the eighteenth century. The objectification of individuals became the means for their subjection and the subjection of individuals became the means for their objectification (*ibid.*: 224). Through the diffusion of psychology and psychiatry, the examination became incarnated in “tests, interviews, interrogations and consultations” that reproduced mutually constitutive power-knowledge relations within disciplinary institutions (*ibid.*: 226–7).

Foucault finds the “formula” for the generalization of the exercise of disciplinary power in Jeremy Bentham’s architectural plan for the model prison, *Panopticon*, published in 1791 (Foucault 2006a: 41). Foucault relates that Bentham depicts the Panopticon as an annular building with an internal periphery consisting of cells containing iron grate doors opening to the interior and windows opening to the exterior as well as a multi-floored central tower containing wide windows with blinds and partitions. Foucault considers this building *the* perfect expression of disciplinary power for a host of reasons. First, with each of the cells designed to be occupied by only one inmate at a time the building produces individualizing effects at its periphery. Second, venetian blinds and partitions on the tower conceal whether anyone actually occupies it, guaranteeing anonymity at the centre. Third, the artificial light from the central tower as well as the natural light entering through the cell windows assure the visibility of inmates in the cells. Finally, this visibility allows for the perpetual writing about inmates and, consequently, the constitution of an administrative knowledge about them (*ibid.*: 75–8).

These features render the Panopticon a magnificent machine not only for subjection but also for *self*-subjection. By inducing *in* inmates an awareness of their own constant visibility, the Panopticon compels them to structure their own behaviour in accordance with its power mechanism (Foucault 1979: 201). Notably missing from this ideal process is any reliance on violence or ostentatious displays of force. Remarkably, the play of visibility facilitated by spatial arrangements and lighting suffices to make inmates the very *conduits* of the power mechanism embodied in the Panopticon.

Though Bentham conceived of the Panopticon as an ideal prison for the resolution of the vexing problem of pauperism (Polanyi 2001: 111–13), Foucault does not tire of reminding us that Bentham considered it applicable to a broad array of settings *besides* the prison. As Foucault explains on Bentham’s behalf, “Whenever one is dealing with a multiplicity of individuals on whom a task or a particular form of behavior must be imposed, the panoptic schema may be used” (1979: 205). Moreover, lest we think that the Panopticon simply remained a product of Bentham’s imagination, Foucault points out that, “In the 1830s, the Panopticon became the architectural program of most prison projects” (*ibid.*: 249) and that institutions apart from the prison adopted its architectural dispositions for a wide variety of purposes. As an example of this adoption, Foucault details all of the Panoptic features of the architecture of the asylum in the early nineteenth century, demonstrating that the panoptic architecture of the asylum building was construed as the very cure to madness (2006a: 102–7). This cross-institutional takeover of Panoptic architectural dispositions intensified the spread of disciplinary power.

For all of the reasons elaborated above, namely, the diffusion of disciplinary power from one institution to another as well as its various transformations into an ever more productive and pervasive modality of power culminating in the extension of Panoptic architectural features, Foucault finds warrant in speaking somewhat grandiosely about the advent of a “disciplinary society”. Yet his employment of this expression is not without qualification. Foucault clearly wants us to take away from the phrase “disciplinary society” an understanding of a society in which disciplinary power is pervasive enough to interact with and alter other modalities of power rather than one in which it simply effaces these other modalities (1979: 216). Such complex articulations derive precisely from the *incompleteness* of the exercise of disciplinary power *even* in the context of a “disciplinary society”. This incompleteness will become abundantly evident as we turn to Taylor’s *Principles*.

Taylor’s Principles as a disciplinary programme

At its core, scientific management as propounded by Taylor in his *Principles* attempts to increase the efficiency of workers by divesting them of any roles in planning and controlling their own work, and by placing these roles squarely in the hands of the management. Scientific management is manifestly disciplinary in this overall goal of increasing efficiency. However, Taylor devotes the bulk of his *Principles* to illustrating the efficacy and superiority of scientific management with reference to concrete examples drawn from a range of industrial activities, and it is within the inglorious intricacies of these examples that his espousal of a disciplinary perspective becomes altogether striking. Let us turn to a couple of his most pertinent illustrations.

Taylor’s first illustration comes from his experience of attempting to increase the amount of pig iron loaded at the Bethlehem Steel Company from 12.5 tons to 47 tons per worker per day. Taylor recounts that he sought this nearly fourfold increase from workers without at the same time provoking their resistance. In his words, “It was further our duty to see that this work was done without bringing on a strike among men” (F. W Taylor 1967: 43). Taylor addresses here the disciplinary problem of maximizing the utility as well as docility of individual bodies. He goes on to explain that he set about resolving this problem by, first of all, selecting out of the 75 workers at Bethlehem Steel four workers capable of loading 47 tons of pig iron per day. This selection took place on the basis of the deployment of a veritable myriad of disciplinary practices, which Taylor describes in the following passage:

In dealing with workmen under this type of management, it is an inflexible rule to talk to and deal with only one man at a time, since each workman has his own special abilities and limitations, and since we are not dealing with men in masses, but are trying to develop each individual man to his highest state of efficiency and prosperity. Our first step was therefore to find a proper workman to begin with. We therefore carefully watched and studied these 75 men for three or four days, at the end of which time we had picked out four men who appeared to be physically able to handle pig iron at the rate of 47 tons per day. A careful study was then made of each of these men. We looked up their history as far back as practicable and thorough inquiries were made as to the character, habits and the ambition of each of them. Finally we selected one from among the four as the most likely man to start with.

(*Ibid.*)

One of the most obvious disciplinary effects in this passage is individualization. Taylor informs us of the importance of having treated workers individually rather than collectively, thereby reminding us of Foucault's discussion of the constitution of a "cellular" individuality. However, he proceeds to indicate that the individualizing effects sought in the selection process rested on the observation of the mass of workers, and that this observation and the knowledge obtained from it facilitated a judgement about the most able-bodied workers. Lastly, Taylor tells us that he made the selection of the most able-bodied worker at Bethlehem Steel on the basis of an enquiry into the identities of the four most able-bodied workers. Individualization, observation and the constitution of an administrative identity on the basis of knowledge obtained through observation all figure centrally in Taylor's account of the selection of the appropriate worker to load 47 tons of pig iron per day.

This worker turned out to be "a little Pennsylvania Dutchman" dubbed Schmidt (*ibid.*). Taylor explains that his team selected Schmidt as the first worker to try out the increase in pig iron loading because it had learned through its enquiries that Schmidt placed an unusually high premium on his earnings. It was on the basis on this knowledge about Schmidt that Taylor's team approached him, first to entice him with the monetary incentive of a pay of \$1.85 rather than standard \$1.15 a day and then to inform him that the increase in pay would presuppose a strict obedience (*ibid.*: 44–5). Taylor relates that Schmidt was told the following in particular:

Well, if you are a high-priced man, you will do exactly as this man tells you to-morrow, from morning 'til night. When he tells you to pick up a pig and walk, you pick it up and you walk, and when he tells you to sit down and rest, you sit down. You do that right straight through the day. And what's more, no back talk. Now a high-priced man does just what he's told to do, and no back talk. Do you understand that? When this man tells you to walk, you walk; when he tells you to sit down, you sit down, and you don't talk back at him.

(*Ibid.*: 45–6)

As in the case of the previous passage, several disciplinary practices leap out at us from this excerpt of "rough talk" to Schmidt (*ibid.*: 46). The first of these practices is the exhaustive regularity of the movements of the body. The person in charge of Schmidt would command him not only how to work but also when and how to rest so as to work all the more efficiently. Moreover, this person would insist that Schmidt follow his orders without any "back talk", once again illustrating the disciplinary relationship between increased utility and increased obedience. In this instance, we are further reminded of Foucault's contention that commands in the exercise of disciplinary power need not be premised on *any* explanation. They need only "trigger off the required behavior and that is enough" (1979: 166).

We learn from Taylor's narrative that Schmidt accepted the conditions spelled out in the passage above and succeeded in loading 47 tons of pig iron per day under the meticulous control of the aforementioned person from Bethlehem Steel. Presumably, Schmidt could have used his demonstrated skill in loading so much pig iron to extract concessions from management at Bethlehem Steel. However, Taylor adds that under the continued presence of an overseer Schmidt "practically never failed to work at this pace and do the task that was set him during the three years that the writer was at Bethlehem" (1967: 47). He thus leaves us with the distinct impression that the application of scientific management succeeded in yielding Schmidt as a docile as well as useful individual.

Another illustration that reveals the disciplinary character of scientific management derives from Taylor's account of efforts to increase the output of shovellers at Bethlehem Steel. Unlike the example above, Taylor in this instance discloses the process used to determine the appropriate load for the maximum daily output of shovellers, explaining that experimentation with and the observation of several of the best shovellers led to the discovery of 21 pounds as the appropriate load (*ibid.*: 65). On the basis of this knowledge, Bethlehem Steel set up an intricate instructions and record-keeping procedure. Taylor recalls the emphasis on individualization in this procedure, stressing that this emphasis was reflected in a system of writing detailing not only what each worker should do on a given day and how he should do it but also whether he had worked enough to earn his \$1.85 on the *previous* day, presumably by loading his 21 pounds per shovel. This system of writing used a norm – work equivalent to \$1.85 a day or 21 pounds per shovel – to distinguish normal from abnormal workers, with the former denoted by the receipt of white slips and the latter denoted by the receipt of yellow slips. According to Taylor, the consequences of this distinction were perfectly clear. Workers in receipt of yellow slips were faced with the choice of working in adherence to the norm and, consequently, normalizing themselves, or being demoted to a type of work corresponding to their comparatively low levels of productivity. We have in his discussion of the consequences of the distribution of these slips a fairly clear and vivid illustration of normalizing judgement.

Taylor draws the following lesson from the experience of individualized instructions and record keeping at Bethlehem Steel: This prescription is most obviously disciplinary in its preference for first training abnormal workers on an individual basis rather than simply discharging them. Training in this instance also facilitates an intimate examination of the aptitudes of these workers with the effect of allowing for the production of additional knowledge about them.

If the workman fails to do his task, some competent teacher should be sent to show him exactly how his work can best be done, to guide, help, and encourage him, and, at the same time, to study his possibilities as a workman. So that, under the plan which individualizes each workman, instead of brutally discharging the man or lowering his wages for failing to make good at once, he is given the time and the help required to make him proficient at his present job, or he is shifted to another class of work for which he is either mentally or physically better suited.

(*Ibid.*: 69–70)

Taylor identifies the point of support for such prescriptions as a ramified structure consisting as it did at Bethlehem Steel of superintend-ants and clerks planning the fine details of work, preparing instruction slips and managing records, subtended by teachers working intimately with workers to make sure that they carry out the tasks spelled out in the instruction slips as well as tool-room men preparing standardized implements for the execution of these tasks (*ibid.*). This structure nicely illustrates the network of gazes facilitating the play of hierarchical observation.

Taylor's exposition of scientific management abounds with such examples of disciplinary power but it also de-naturalizes this modality of power by demonstrating that disciplinary practices bound up with the application of scientific management have yet to take root, and that they are subject to great contestation (at the risk of belabouring the obvious, it is worth keeping in mind that if these disciplinary practices *had* taken root, Taylor would not have needed to compose *Principles*). Indeed, Taylor paints his vision of

scientifically managed labour-processes on a canvas of sceptical employers as well as a mass of recalcitrant, if not openly hostile, workers. Against the disciplinary demand for maximized utility, we find in Taylor's narrative workers who threaten to strike, workers who intimidate managers and workers who damage machinery (*ibid.*: 49–52). Taylor is acutely aware of the prospect of inciting warfare within the interstices of the labour process. For this reason, he warns prospective practitioners against the hasty application of his principles of scientific management and insists that only a prolonged period of habituation will result in the successful application of these principles (*ibid.*: 128–35). To paraphrase Foucault, we could therefore say that the not so distant “roar of battle” (Foucault 1979: 308) resounds throughout Taylor's theory, suggesting that *only* an ensemble of deeply contested practices sustain the exercise of a disciplinary power that strives to appear natural and spontaneous in the very bodies of individuals.